

1. (a)

If given points are collinear, then

$$\frac{x_1 - x_2}{x_2 - x_3} = \frac{y_1 - y_2}{y_2 - y_3} = \frac{z_1 - z_2}{z_2 - z_3}$$

$$\Rightarrow \frac{5-6}{6-8} = \frac{2+1}{-1+7} = \frac{4-2}{2-k}$$

$$\Rightarrow \frac{-1}{-2} = \frac{3}{6} = \frac{2}{2-k} \Rightarrow \frac{1}{2} = \frac{2}{2-k}$$

$$\Rightarrow k = -2$$

2. (b)

We have, $\mathbf{r} = (4\mathbf{i} - \mathbf{j}) + s(2\mathbf{i} + \mathbf{j} - 3\mathbf{k})$ and

$$\mathbf{r} = (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) + t(\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})$$

$$\cos \theta = \frac{\mathbf{b}_1 \cdot \mathbf{b}_2}{|\mathbf{b}_1| |\mathbf{b}_2|}$$

We know that,

$$\cos \theta = \frac{(2\mathbf{i} + \mathbf{j} - 3\mathbf{k}) \cdot (\mathbf{i} - 3\mathbf{j} + 2\mathbf{k})}{\sqrt{4+1+9}\sqrt{1+9+4}} = \frac{2-3-6}{\sqrt{14} \cdot \sqrt{14}} = \frac{-7}{14}, \quad \cos \theta = -\frac{1}{2}$$

$$\theta = \cos^{-1}\left(\frac{1}{2}\right) \quad \text{i.e.} \quad \theta = \frac{\pi}{3}$$

Hence, acute angle

3. (a)

$$\text{Equation of BC, } \frac{x-4}{-1} = \frac{y-7}{-2} = \frac{z-1}{2}$$

$$\text{i.e. } \frac{x-4}{1} = \frac{y-7}{2} = \frac{z-1}{-2} = r \quad (\text{say})$$

Any point on the given line is $D(r+4, 2r+7, -2r+1)$

Then, d.r.'s of $AD = (r+4-1, 2r+7-0, -2r+1-3)$

i.e. d.r.'s of $AD = (r+3, 2r+7, -2r-2)$ and

d.r.'s of $BC = (-1, -2, 2)$

Since AD is $\perp$ to given line,

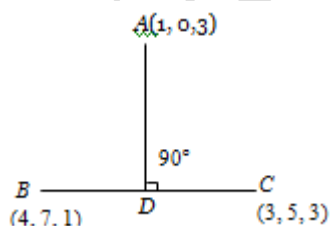
$$\therefore (-1)(r+3) + (2r+7)(-2) + (2)(-2r-2) = 0$$

$$\Rightarrow -r-3-4r-14-4r-4 = 0 \Rightarrow -9r-21 = 0$$

$$\Rightarrow r = -7/3$$

$$\therefore D \text{ is } \{4 - (7/3), 7 - (14/3), (14/3) + 1\}$$

$$\text{i.e. } D \text{ is } (5/3, 7/3, 17/3)$$



4. (d)

D.R's of $\vec{CD}$ are $(2, -2, -1)$ or $(-2, 2, 1)$

$$\therefore \text{D.C's of } \vec{CD} \text{ are } \left(\frac{-2}{3}, \frac{2}{3}, \frac{1}{3}\right)$$

Projection of $\vec{AB}$ on

$$CD = (6-1)\left(-\frac{2}{3}\right) + (7-2)\left(\frac{2}{3}\right) + (8-3)\left(\frac{1}{3}\right)$$

$$= 5 \left(\frac{-2+2+1}{3} \right) = \frac{5}{3}$$

5. (d)

Equation of line AB : $\frac{x-1}{1} = \frac{y-0}{3} = \frac{z+3}{z}$

$\Rightarrow$ Any point on the line is $(\lambda + 1, 3\lambda, 2\lambda - 3)$

It lies on the plane

So $(\lambda + 1) + 2(3\lambda) - 3(2\lambda - 3) = 12$

$\Rightarrow \lambda = 2$

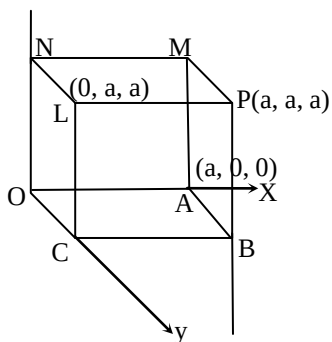
$\Rightarrow P(3, 6, 1)$

$\therefore AP^2 = 4 + 36 + 16 = 56$

6. (b)

dr's of AL $a, -a, -a$

dr's of AP a, a, a



θ between AL and AP

$$\cos \theta = \frac{a \cdot 0 + a(-a) + a(-a)}{\sqrt{a^2 + (-a)^2 + (-a)^2} \sqrt{0^2 + a^2 + a^2}} = -\frac{\sqrt{2}}{\sqrt{3}}$$

$\Rightarrow \tan \theta = \frac{1}{\sqrt{2}}$

7. (b)

Here $a = \sqrt{4+4+1} = \sqrt{9}$, $b = \sqrt{1+4+4} = \sqrt{9}$ and

$c = \sqrt{1+16+1} = \sqrt{18}$.

Obviously it is a right angled and isosceles triangle.

8. (b)

$x = \frac{-5+9}{-2} = -2$, $y = \frac{-5(2)+3(-5)}{-2} = \frac{25}{2}$

$z = \frac{-5(3)+3(6)}{-2} = -\frac{3}{2}$.

9. (b)

Let $A(1, 1, 1)$, $B(-2, 4, 1)$, $C(-1, 5, 5)$ & $D(2, 2, 5)$

$AB = \sqrt{9+9+0} = 3\sqrt{2}$, $BC = \sqrt{1+1+16} = 3\sqrt{2}$ and

$CD = 3\sqrt{2}$ and $AD = 3\sqrt{2}$. Hence it is a square.

10. (b)

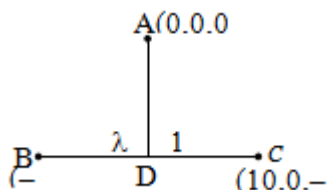
As $\frac{a}{(1/bc)} = \frac{b}{(1/ca)} = \frac{c}{(1/ab)}$, Hence lines are parallel.

11. (d)

Let AD be perpendicular and D be foot of perpendicular which divide BC in ratio $\lambda : 1$, then

$$D\left(\frac{10\lambda-9}{\lambda+1}, \frac{4}{\lambda+1}, \frac{-\lambda+5}{\lambda+1}\right)$$

.....(i)



The direction ratio of AD are $\frac{10\lambda-9}{\lambda+1}, \frac{4}{\lambda+1}, \frac{-\lambda+5}{\lambda+1}$ and direction ratio of BC are 19, -4 and -6.

Since $AD \perp BC$

$$\Rightarrow 19\left(\frac{10\lambda-9}{\lambda+1}\right) - 4\left(\frac{4}{\lambda+1}\right) - 6\left(\frac{-\lambda+5}{\lambda+1}\right) = 0$$

$$\Rightarrow \lambda = \frac{31}{28}$$

Hence on putting the value of λ in (i), we get required foot of the perpendicular i.e., $\left(\frac{58}{59}, \frac{112}{59}, \frac{109}{59}\right)$.

Trick: The line passing through these points is $\frac{x+9}{19} = \frac{y-4}{-4} = \frac{z-5}{-6}$. Now co-ordinates of the foot lie on this line, so they must satisfy the given line. But here no point satisfies the line, hence answer is (d).

12. (a)

$$\text{Required distance} = \left| \frac{6-18+8+11}{7} \right| = 1$$

13. (b)

It is a line passing through (α, β, γ) and whose direction cosines are a, b, c .

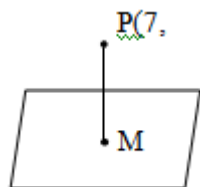
14. (d)

Let M be the foot of perpendicular from (7, 14, 5) to the given plane, then PM is normal to the plane. So, its d.r.'s are 2, 4, -1.

Since PM passes through $P(7, 14, 5)$ and has d.r.'s 2, 4, -1.

$$\frac{x-7}{2} = \frac{y-14}{4} = \frac{z-5}{-1} = r$$

Therefore, its equation is



(Say)

$$\Rightarrow x = 2r+7, y = 4r+14, z = -r+5$$

Let co-ordinates of M be $(2r+7, 4r+14, -r+5)$

Since M lies on the plane $2x+4y-z=2$, therefore $2(2r+7)+4(4r+14)-(-r+5)=2 \Rightarrow r=-3$

So, co-ordinates of foot of perpendicular are $M(1, 2, 8)$

Now, PM = Length of perpendicular from P

$$= \sqrt{(1-7)^2 + (2-14)^2 + (8-5)^2} = 3\sqrt{21}$$

15. (a)

Let the co-ordinates of the points where the plane cuts the axes are $(a, 0, 0), (0, b, 0), (0, 0, c)$. Since centroid is (α, β, γ) , therefore $a=3\alpha, b=3\beta, c=3\gamma$.

$$\text{Equation of the plane will be } \frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$$

$$\Rightarrow \frac{x}{3\alpha} + \frac{y}{3\beta} + \frac{z}{3\gamma} = 1 \Rightarrow \frac{x}{\alpha} + \frac{y}{\beta} + \frac{z}{\gamma} = 3.$$

16. (b)

Equation of a plane, when direction ratio and length of perpendicular is given, $ax + by + cz = p \sqrt{a^2 + b^2 + c^2}$

Given, $(a, b, c) \rightarrow (-3, 2, 6)$

$$-3x + 2y + 6z = 7\sqrt{(-3)^2 + 2^2 + 6^2}$$

$$-3x + 2y + 6z = 49.$$

17. (b)

$$x^2 - 5x + 6 = 0 \Rightarrow (x - 2) = 0 \text{ or } (x - 3) = 0,$$

which represents a plane.

18. (b)

According to question, $|3 + 4 - 12k + 13| = |-9 - 12 + 13|$

$$\therefore 3 + 4 - 12k + 13 = 8 \Rightarrow k = 1.$$

19. (c)

According to question, $4x - 7y + 3z = k$ (i)

Also, plane (i) passes through $\left(\frac{-1+3}{2}, \frac{2-5}{2}, \frac{3+6}{2}\right)$, then $4 + \frac{21}{2} + \frac{27}{2} = k \Rightarrow k = 28$

Therefore, required equation is $4x - 7y + 3z = 28$.

20. (d)

Obviously the line and the plane are parallel, so to find the distance between the line and the plane, take any point on the line i.e., $(1, -2, 1)$. Now the perpendicular distance of the point $(1, -2, 1)$ from the plane will be the required distance.

$$= \left| \frac{2(1) + 2(-2) - 1(1) - 6}{\sqrt{2^2 + 2^2 + 1^2}} \right| = \frac{9}{\sqrt{9}} = 3$$

Hence distance